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Heat and Thermodynamics

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	93 questions
Answer key	at the end

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MCQ (Single Correct Answer) 93 questions

Q1. RE-NEET 2026

The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is:

Open solution

(A) $6\pi R^3\alpha\Delta T$ (B) $2\pi R^3\alpha\Delta T$ (C) $3\pi R^3\alpha\Delta T$ (D) $4\pi R^3\alpha\Delta T$

Q2. RE-NEET 2026

In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma = 5/3$) decreases from 60 K to 50 K . The work done by the gas in the process is:

(Take the universal gas constant as $R = 8.3\ \text{J mol}^{-1}\ \text{K}^{-1}$)

Open solution

(A) 166 J (B) 41.5 J (C) 83 J (D) 124.5 J

Q3. RE-NEET 2026

An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_P/C_V of the gas is $8/7$, then the value of f is:

Open solution

(A) 1 (B) 4 (C) 3 (D) 2

Q4. RE-NEET 2026

The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, the correct option is:

Open solution

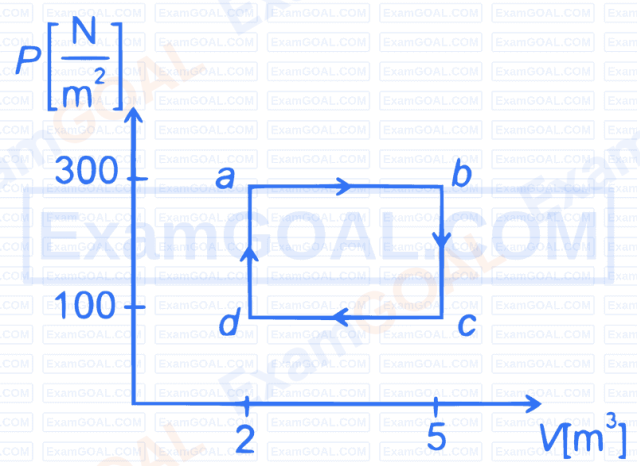
(A) $n_A = \frac{1}{2}n_B$ (B) $n_A = n_B$ (C) $n_A = 2n_B$ (D) $n_A = \frac{1}{4}n_B$

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Q5. RE-NEET 2026

One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is:

Open solution



(A) 800 J (B) 400 J (C) 500 J (D) 600 J

Q6. NEET 2026

A flask contains argon and chlorine in the ratio of $2:1$ by mass. The temperature of the mixture is $27^{\circ}\mathrm{C}$. The ratio of root mean square speed of the molecules of the two gases $\left(\frac{V_{\mathrm{rms}}^{\mathrm{Ar}}}{V_{\mathrm{rms}}^{\mathrm{Cl}}}\right)$ is:

(Atomic mass of argon $= 40.0\mathrm{u}$ and molecular mass of chlorine $= 70.0\mathrm{u}$)

Open solution

(A) $\frac{\sqrt{7}}{2}$ (B) $\frac{7}{4}$ (C) $\frac{7}{2}$ (D) $\frac{2}{\sqrt{7}}$

Q7. NEET 2026

An electric heater supplies heat to a system at a rate of 100 W . If the system performs work at a rate of $75\ \mathrm{J/s}$, then the rate at which internal energy increases will be:

Open solution

(A) 75 W (B) 100 W (C) 125 W (D) 25 W

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Q8. NEET 2025

Open solution

Three identical heat conducting rods are connected in series as shown in the figure. The rods on the sides have thermal conductivity $2K$ while that in the middle has thermal conductivity K. The left end of the combination is maintained at temperature $3T$ and the right end at T. The rods are thermally insulated from outside. In steady state, temperature at the left junction is T_1 and that at the right junction is T_2. The ratio T_1/T_2 is



(A) $\frac{5}{3}$ (B) $\frac{5}{4}$ (C) $\frac{3}{2}$ (D) $\frac{4}{3}$

Q9. NEET 2025

Open solution

A container has two chambers of volumes $V_1 = 2$ litres and $V_2 = 3$ litres separated by a partition made of a thermal insulator. The chambers contain $n_1 = 5$ and $n_2 = 4$ moles of ideal gas at pressures $p_1 = 1\ \text{atm}$ and $p_2 = 2\ \text{atm}$, respectively. When the partition is removed, the mixture attains an equilibrium pressure of

(A) 1.4 atm (B) 1.8 atm (C) 1.3 atm (D) 1.6 atm

Q10. NEET 2025

Open solution

An oxygen cylinder of volume 30 litre has 18.20 moles of oxygen. After some oxygen is withdrawn from the cylinder, its gauge pressure drops to 11 atmospheric pressures at temperature 27°C. The mass of the oxygen withdrawn from the cylinder is nearly equal to:

[Given, $R = \frac{100}{12}\ \text{J mol}^{-1}\ \text{K}^{-1}$, and molecular mass of $\text{O}_2 = 32$, 1 atm pressure $= 1.01 \times 10^5\ \text{N/m}^2$]

(A) 0.116 kg (B) 0.156 kg (C) 0.125 kg (D) 0.144 kg

Q11. NEET 2025

Open solution

Two gases A and B are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B, respectively. On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas A and B are displaced by 16 cm and 9 cm , respectively. If the change in their internal energy is the same, then the ratio $\frac{r_A}{r_B}$ is equal to

(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{4}{3}$ (D) $\frac{3}{4}$

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Q12. NEET 2024 (Re-Examination)

Open solution

Given below are two statements: One is labelled as Assertion **A** and the other is labelled as Reason **R**.

Assertion A: Houses made of concrete roofs overlaid with foam keep the room hotter during summer.

Reason R: The layer of foam insulation prohibits heat transfer, as it contains air pockets.

In the light of the above statements, choose the correct answer from the options given below.

(A) A is true but R is false.

(B) A is false but R is true.

(C) Both A and R are true and R is the correct explanation of A.

(D) Both A and R are true but R is NOT the correct explanation of A.

Q13. NEET 2024 (Re-Examination)

Open solution

The equilibrium state of a thermodynamic system is described by

A. Pressure

B. Total heat

C. Temperature

D. Volume

E. Work done

Choose the most appropriate answer from the options given below.

(A) A, B and E only **(B)** B, C and D only **(C)** A, B and C only **(D)** A, C and D only

Q14. NEET 2024 (Re-Examination)

Open solution

According to the law of equipartition of energy, the number of vibrational modes of a polyatomic gas of constant $\gamma = \frac{C_p}{C_v}$ is (C_P and C_V are the specific heat capacities of the gas at constant pressure and constant volume, respectively):

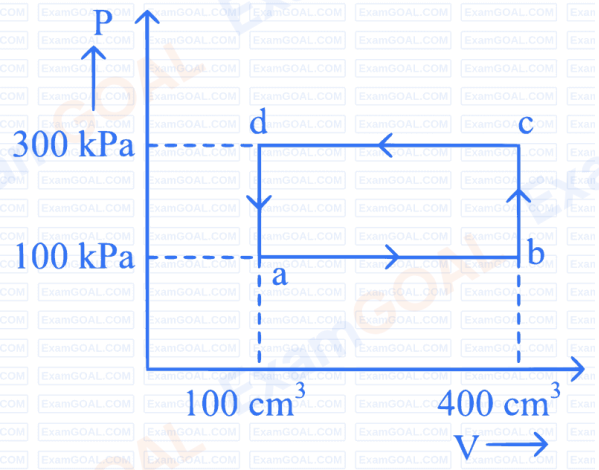
(A) $\frac{4+3\gamma}{\gamma-1}$ **(B)** $\frac{3+4\gamma}{\gamma-1}$ **(C)** $\frac{4-3\gamma}{\gamma-1}$ **(D)** $\frac{3-4\gamma}{\gamma-1}$

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Q15. NEET 2024

A thermodynamic system is taken through the cycle $abcda$. The work done by the gas along the path bc is:

Open solution

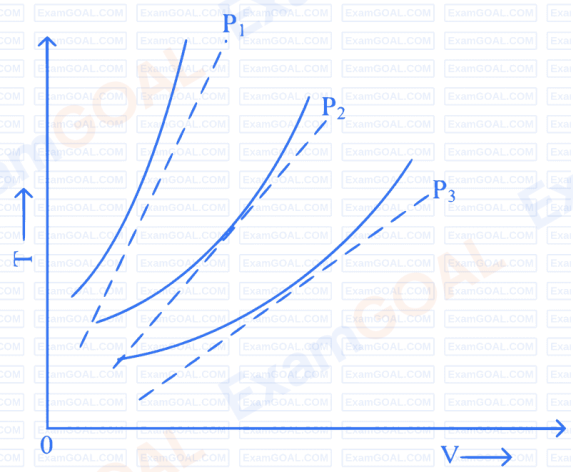


(A) Zero (B) 30 J (C) -90 J (D) -60 J

Q16. NEET 2024

The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1, P_2 and P_3 compared with those of Charles's law represented as dotted lines.

Open solution



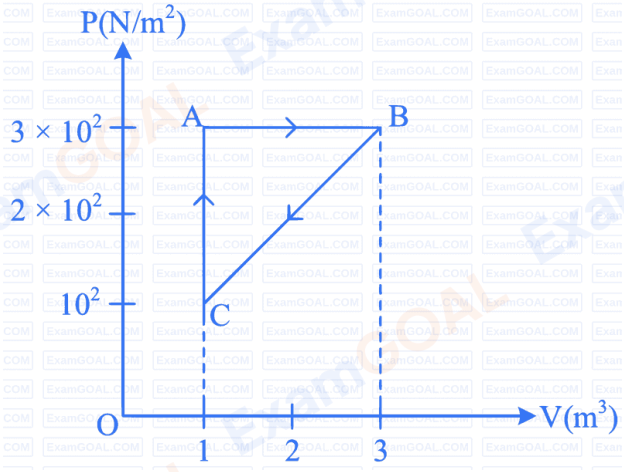
Then the correct relation is :

(A) $P_3 > P_2 > P_1$ (B) $P_1 > P_3 > P_2$ (C) $P_2 > P_1 > P_3$ (D) $P_1 > P_2 > P_3$

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Q17. NEET 2023 Manipur

For the given cycle, the work done during isobaric process is:



Open solution

(A) 200 J (B) Zero (C) 400 J (D) 600 J

Q18. NEET 2023 Manipur

A container of volume $200\ \text{cm}^3$ contains 0.2 mole of hydrogen gas and 0.3 mole of argon gas. The pressure of the system at temperature $200\ \text{K}$ ($\text{R} = 8.3\text{JK}^{-1}\ \text{mol}^{-1}$) will be :-

(A) $6.15 \times 10^5\ \text{Pa}$ (B) $6.15 \times 10^4\ \text{Pa}$ (C) $4.15 \times 10^5\ \text{Pa}$ (D) $4.15 \times 10^6\ \text{Pa}$

Open solution

Q19. NEET 2023

The temperature of a gas is -50°C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?

(A) 3295°C (B) $3097\ \text{K}$ (C) $223\ \text{K}$ (D) 669°C

Open solution

Q20. NEET 2023

A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C. The temperature of the sink is :-

(A) 15°C (B) 100°C (C) 200°C (D) 27°C

Open solution

Q21. NEET 2022 Phase 2

An ideal gas follows a process described by the equation $PV^2 = C$ from the initial (P_1, V_1, T_1) to final (P_2, V_2, T_2) thermodynamic states, where C is a constant. Then

(A) If $P_1 > P_2$ then $V_1 > V_2$ (B) If $P_1 > P_2$ then $T_1 < T_2$

(C) If $V_2 > V_1$ then $T_2 > T_1$ (D) If $V_2 > V_1$ then $T_2 < T_1$

Open solution

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Q22. NEET 2022 Phase 2

Two rods one made of copper and other made of steel of same length and same cross sectional area are joined together. The thermal conductivity of copper and steel are 385 J s^{-1} K^{-1} m^{-1} and 50 J s^{-1} K^{-1} m^{-1} respectively. The free ends of copper and steel are held at 100°C and 0°C respectively. The temperature at the junction is, nearly :

Open solution

(A) $88.5^{\circ}C$ (B) $12^{\circ}C$ (C) $50^{\circ}C$ (D) $73^{\circ}C$

Q23. NEET 2022 Phase 2

Three vessels of equal capacity have gases at the same temperature and pressure. The first vessel contains helium (monoatomic), the second contains fluorine (diatomic) and the third contains sulfur hexafluoride (polyatomic). The correct statement, among the following is :

Open solution

(A) The root mean square speed of sulfur hexafluoride is the largest

(B) All vessels contain unequal number of respective molecules

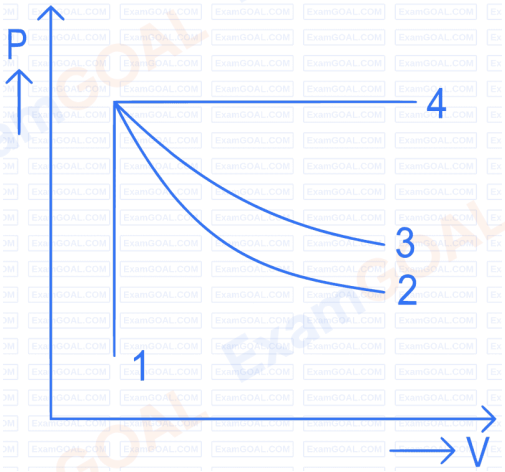
(C) The root mean square speed of molecules is same in all three cases

(D) The root mean square speed of helium is the largest

Q24. NEET 2022 Phase 1

An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal, isobaric and isochoric. The curve which represents the adiabatic process among 1, 2, 3 and 4 is

Open solution



(A) 1 (B) 2 (C) 3 (D) 4

Q25. NEET 2022 Phase 1

The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces vanish away is

Open solution

(A) 5.6×10^{6} m^3 (B) 5.6×10^{3} m^3 (C) 5.6×10^{-3} m^3 (D) 5.6 m^3

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Q26. NEET 2021

A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C. The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is :

(A) $\frac{5}{13}$t (B) $\frac{13}{10}$t (C) $\frac{13}{5}$t (D) $\frac{10}{13}$t

Open solution

Q27. NEET 2021

Match Column - I and Column - II and choose the correct match from the given choices.

Open solution

	Column - I		Column - II
(A)	Root mean square speed of gas molecules	(P)	$\frac{1}{3}nm\bar{v}^2$
(B)	Pressure exerted by ideal gas	(Q)	$\sqrt{\frac{3RT}{M}}$
(C)	Average kinetic energy of a molecule	(R)	$\frac{5}{2}RT$
(D)	Total internal energy of 1 mole of a diatomic gas	(S)	$\frac{3}{2}k_BT$

(A) (A) - (R), (B) - (Q), (C) - (P), (D) - (S)

(B) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)

(C) (A) - (Q), (B) - (R), (C) - (S), (D) - (P)

(D) (A) - (Q), (B) - (P), (C) - (S), (D) - (R)

Q28. NEET 2020 Phase 1

The average thermal energy for a mono-atomic gas is : (k_B is Boltzmann constant and T absolute temperature)

(A) $\frac{3}{2}k_BT$ (B) $\frac{5}{2}k_BT$ (C) $\frac{7}{2}k_BT$ (D) $\frac{1}{2}k_BT$

Open solution

Q29. NEET 2020 Phase 1

The mean free path for a gas, with molecular diameter d and number density n can be expressed as :

(A) $\frac{1}{\sqrt{2}n\pi d^2}$ (B) $\frac{1}{\sqrt{2}n^2\pi d^2}$

(C) $\frac{1}{\sqrt{2}n^2\pi^2 d^2}$ (D) $\frac{1}{\sqrt{2}n\pi d}$

Open solution

Q30. NEET 2020 Phase 1

A cylinder contains hydrogen gas at pressure 249 kPa and temperature 27°C

Its density is : (R = 8.3 J mol^{-1} K^{-1})

(A) 0.2 kg/m^3 (B) 0.1 kg/m^3 (C) 0.02 kg/m^3 (D) 0.5 kg/m^3

Open solution

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Q31. NEET 2020 Phase 1

The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5r_2$) through 1 K are in the ratio :

(A) $\frac{9}{4}$ (B) $\frac{3}{2}$ (C) $\frac{5}{3}$ (D) $\frac{27}{8}$

Open solution

Q32. NEET 2020 Phase 1

Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire systems is thermally insulated. The stop cock is suddenly opened. The Process is :

(A) adiabatic (B) isochoric (C) isobaric (D) isothermal

Open solution

Q33. NEET 2019

In which of the following processes, heat is neither absorbed nor released by a system?

(A) adiabatic (B) isobaric (C) isochoric (D) isothermal

Open solution

Q34. NEET 2019

Increase in tempertaure of a gas filled in a container would lead to :

(A) increase in its kinetic energy (B) decrease in intermolecular distance

(C) decrease in its pressure (D) increase in its mass

Open solution

Q35. NEET 2019

A copper rod of 88 cm and an aluminium rod of unknown length have their increase in length independent of increase in temperature. The length of aluminium rod is : ($\alpha_{Cu} = 1.7 \times 10^{-5}\ K^{-1}$ and $\alpha_{Al} = 2.2 \times 10^{-5}\ K^{-1}$)

(A) 119.9 cm (B) 88 cm (C) 68 cm (D) 6.8 cm

Open solution

Q36. NEET 2018

The power radiated by a black body is P and it radiates maximum energy at wavelength, λ_0 . If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it becomes nP. The value of n is

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{256}{81}$ (D) $\frac{81}{256}$

Open solution

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Q37. NEET 2018

A sample of 0.1 g of water at 100°C and normal pressure (1.013×10^5 N m^{-2}) requires 54 cal of heat energy to convert to steam at 100°C. If the volume of the steam produced is 167.1 cc, the change in internal energy of the sample, is

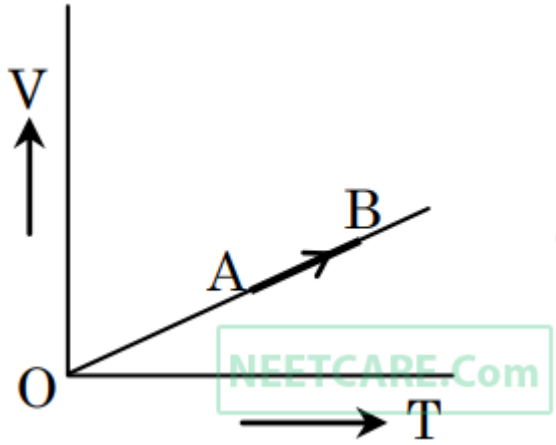
Open solution

(A) 104.3 J (B) 208.7 J (C) 42.2 J (D) 84.5 J

Q38. NEET 2018

The volume (V) of a monatomic gas varies with its temperature (T), as shown in the graph. The ratio of work done by the gas, to the heat absorbed by it, when it undergoes a change from state A to state B, is

Open solution



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(A) $\frac{2}{5}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{2}{7}$

Q39. NEET 2018

The efficiency of an ideal heat engine working between the freezing point and boiling point of water, is

Open solution

(A) 26.8% (B) 20% (C) 6.25% (D) 12.5%

Q40. NEET 2018

At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere?

(Given : Mass of oxygen molecule (m) = 2.76×10^{-26} kg, Boltzmann's constant $k_B = 1.38 \times 10^{-23}$ J K^{-1})

Open solution

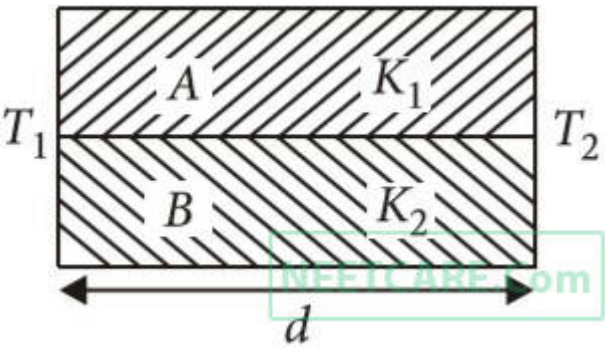
(A) 2.508×10^4 K (B) 8.360×10^4 K (C) 5.016×10^4 K (D) 1.254×10^4 K

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Q41. NEET 2017

Two rods A and B of different materials are welded together as shown in figure. Their thermal conductivities are K_1 and K_2. The thermal conductivity of the composite will be

Open solution



(A) $\frac{3(K_1+K_2)}{2}$

(B) $K_1 + K_2$

(C) $2(K_1 + K_2)$

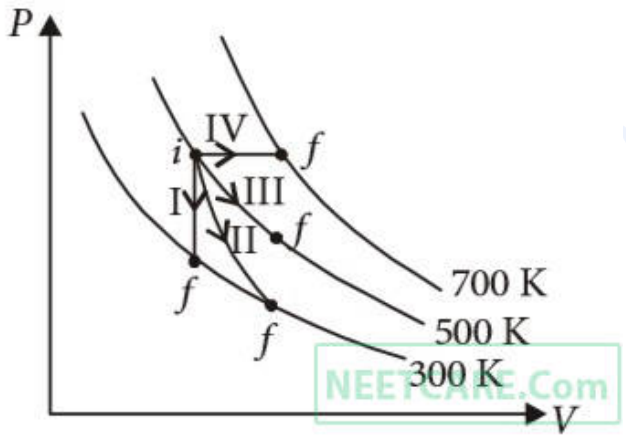
(D) $\frac{K_1+K_2}{2}$

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Q42. NEET 2017

Thermodynamic processes are indicated in the following diagram.

Open solution



Match the following

Column-1			Column-2	
P.	Process I		A.	Adiabatic
Q.	Process II		B.	Isobaric
R.	Process III		C.	Isochoric
S.	Process IV		D.	Isothermal

(A) P $\rightarrow$ C, Q $\rightarrow$ A, R $\rightarrow$ D, S $\rightarrow$ B

(B) P $\rightarrow$ C, Q $\rightarrow$ D, R $\rightarrow$ B, S $\rightarrow$ A

(C) P $\rightarrow$ D, Q $\rightarrow$ B, R $\rightarrow$ A, S $\rightarrow$ C

(D) P $\rightarrow$ A, Q $\rightarrow$ C, R $\rightarrow$ D, S $\rightarrow$ B

Q43. NEET 2017

A carnot engine having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is

(A) 90 J (B) 99 J (C) 100 J (D) 1 J

Open solution

Q44. NEET 2017

A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is

(A) 15 RT (B) 9 RT (C) 11 RT (D) 4 RT

Open solution

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Q45. NEET 2016 Phase 2

One mole of an ideal monatomic gas undergoes a process described by the equation PV^3 = constant. The heat capacity of the gas during this process is

(A) $\frac{3}{2}$ R (B) $\frac{5}{2}$ R (C) 2R (D) R

Open solution

Q46. NEET 2016 Phase 2

The temperature inside a refrigerator is t_2 °C and the room temperature is t_1 °C. The amount of heat delivered to the room for each joule of electrical energy consumed ideally will be

(A) $\frac{t_1}{t_1-t_2}$ (B) $\frac{t_1+273}{t_1-t_2}$
(C) $\frac{t_2+273}{t_1-t_2}$ (D) $\frac{t_1+t_2}{t_1+273}$

Open solution

Q47. NEET 2016 Phase 2

A given sample of an ideal gas occupies a volume V at a pressure P and absolute temperature T. The mass of each molecule of the gas is m. Which of the following gives the density of the gas ?

(A) P/(kT) (B) Pm/(kT) (C) P/(KTV) (D) mkT

Open solution

Q48. NEET 2016 Phase 1

A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then

(A) Compressing the gas isothermally or adiabatically will require the same amount of work.
(B) Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas.
(C) Compressing the gas isothermally will require more work to be done.
(D) Compressing the gas through adiabatic process will require more work to be done.

Open solution

Q49. NEET 2016 Phase 1

The molecules of a given mass of a gas have r.m.s. velocity of 200 m s^{-1} at 27°C and 1.0 $\times$ 10^5 N m^{-2} pressure. When the temperature and pressure of the gas are respectively, 127°C and 0.05 $\times$ 10^5 N m^{-2}, the r.m.s. velocity of its molecules in m s^{-1} is

(A) $\frac{100\sqrt{2}}{3}$ (B) $\frac{100}{3}$ (C) $100\sqrt{2}$ (D) $\frac{400}{\sqrt{3}}$

Open solution

Q50. NEET 2016 Phase 1

A refrigerator works between 4°C and 30°C. It is required to remove 600 calories of heat every second in order to keep the temperature of the refrigerated space constant. The power required is (Take 1 cal = 4.2 Joules)

(A) 236.5 W (B) 2365 W (C) 2.365 W (D) 23.65 W

Open solution

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Q51. AIPMT 2015

An ideal gas is compressed to half its initial volume by means of several processes. Which of the process results in the maximum work done on the gas ?

(A) Isochoric (B) Isothermal (C) Adiabatic (D) Isobaric

Open solution

Q52. AIPMT 2015

Two vessels separately contain two ideal gases A and B at the same temperature, the pressure of A being twice that of B. Under such conditions, the density of A is found to be 1.5 times the density of B. The ratio of molecular weight of A and B is

(A) 2 (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Open solution

Q53. AIPMT 2015

The coefficient of performance of a refrigerator is 5. If the temperature inside freezer is -20°C, the temperature of the surroundings to which it rejects heat is

(A) 11°C (B) 21°C (C) 31°C (D) 41°C

Open solution

Q54. AIPMT 2015 Cancelled Paper

The ratio of the specific heats $\frac{C_p}{C_v} = \gamma$ in terms of degrees of freedom (n) is given by

(A) $\left(1+\frac{2}{n}\right)$ (B) $\left(1+\frac{n}{2}\right)$ (C) $\left(1+\frac{1}{n}\right)$ (D) $\left(1+\frac{n}{3}\right)$

Open solution

Q55. AIPMT 2015 Cancelled Paper

A Carnot engine, having an efficiency of $\eta = \frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is

(A) 90 J (B) 1 J (C) 100 J (D) 99 J

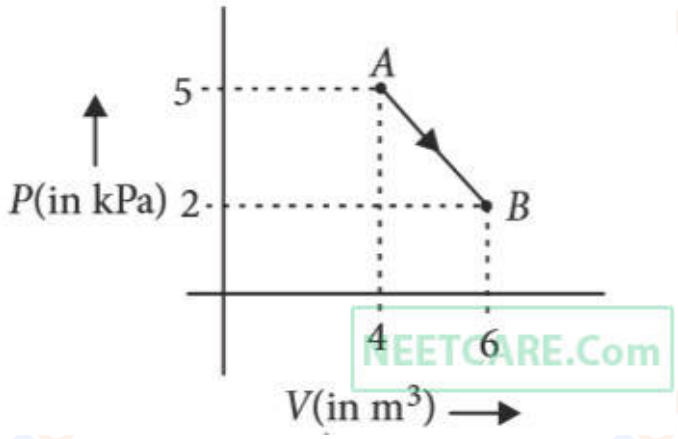
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Q56. AIPMT 2015 Cancelled Paper

Open solution

One mole of an ideal diatomic gas undergoes a transition from A to B along a path AB as shown in the figure.



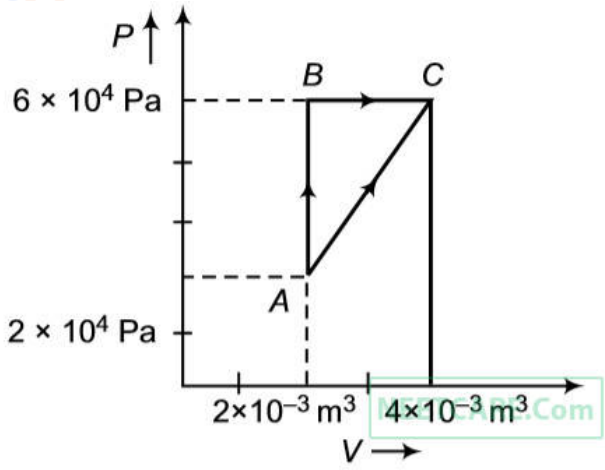
The change in internal energy of the gas during the transition is

(A) 20 J (B) − 12 kJ (C) 20 kJ (D) − 20 kJ

Q57. AIPMT 2015 Cancelled Paper

Open solution

Figure below shows two paths that may be taken by a gas to go from a state A to a state C.



In process AB, 400 J of heat is added to the system and in process BC, 100 J of heat is added to the system. The heat absorbed by the system in the process AC will be

(A) 460 J (B) 300 J (C) 380 J (D) 500 J

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Q58. AIPMT 2014

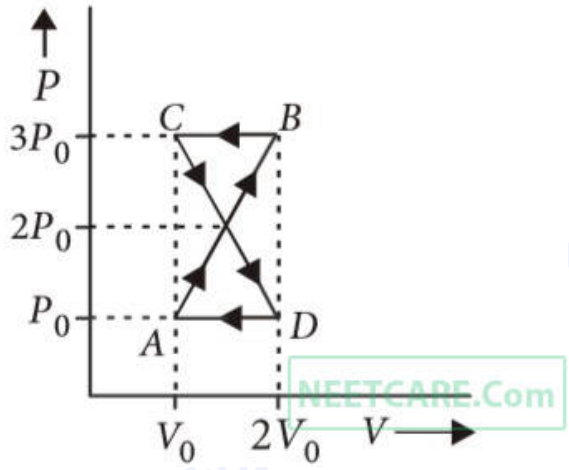
A monatomic gas at a pressure P, having a volume V expands isothermally to a volume 2V and then adiabatically to a volume 16V. The final pressure of the gas is (Take γ = 5/3)

(A) 64P (B) 32P (C) P/64 (D) 16P

Open solution

Q59. AIPMT 2014

A thermodynamic system undergoes cyclic process ABCDA as shown in figure. The work done by the system in the cycle is



(A) P_0V_0 (B) $2P_0V_0$ (C) $\frac{P_0V_0}{2}$ (D) zero

Open solution

Q60. AIPMT 2014

The mean free path of molecules of a gas, (radius r) is inversely proportional to

(A) r^3 (B) r^2 (C) r (D) $\sqrt{r}$

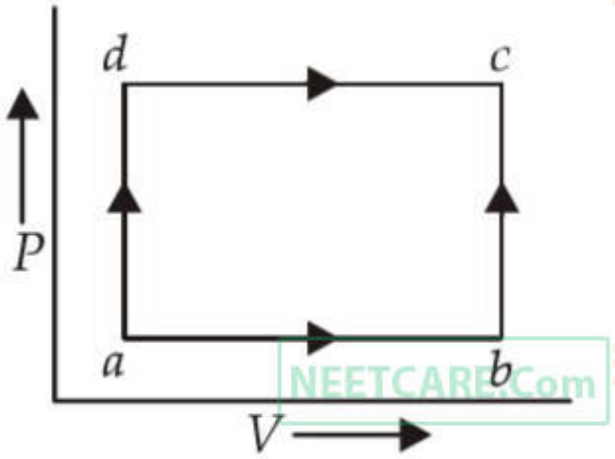
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Q61. NEET 2013 (Karnataka)

A system is taken from state a to state c by two paths adc and abc as shown in the figure. The internal energy at a is $U_a = 10\,J$. Along the path adc the amount of heat absorbed $dQ_1 = 50$ J and the work obtained $dW_1 = 20$ J whereas along the path abc the heat absorbed dQ_2 = 36 J. The amount of work along the path abc is

Open solution



(A) 10 J (B) 12 J (C) 36 J (D) 6 J

Q62. NEET 2013 (Karnataka)

Which of the following relations does not give the equation of an adiabatic process, where terms have their usual meaning?

Open solution

(A) $P^{1-\gamma}T^{\gamma}$ = constant (B) PV^{γ} = constant

(C) $TV^{\gamma-1}$ = constant (D) $P^{\gamma}\,T^{1-\gamma}$ = constant

Q63. NEET 2013 (Karnataka)

Two Carnot engines A and B are operated in series. The engine A receives heat from the source at temperature T_1 and rejects the heat to the sink at temperature T. The second engine B receives the heat at temperature T and rejects to its sink at temperature T_2. For what values of T the efficiencies of the two engines are equal

Open solution

(A) $\frac{T_1-T_2}{2}$ (B) T_1T_2 (C) $\sqrt{T_1T_2}$ (D) $\frac{T_1+T_2}{2}$

Q64. NEET 2013 (Karnataka)

In a vessel, the gas is at pressure P. If the mass of all the molecules is halved and their speed is doubled, then the resultant pressure will be

Open solution

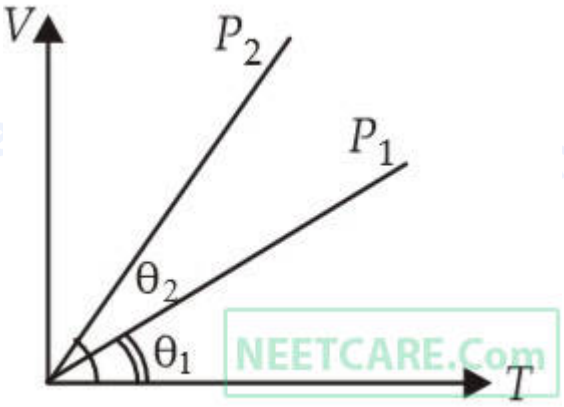
(A) 2P (B) P (C) P/2 (D) 4P

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Q65. NEET 2013

In the given (V – T) diagram, what is the relation between pressure P_1 and P_2?

Open solution

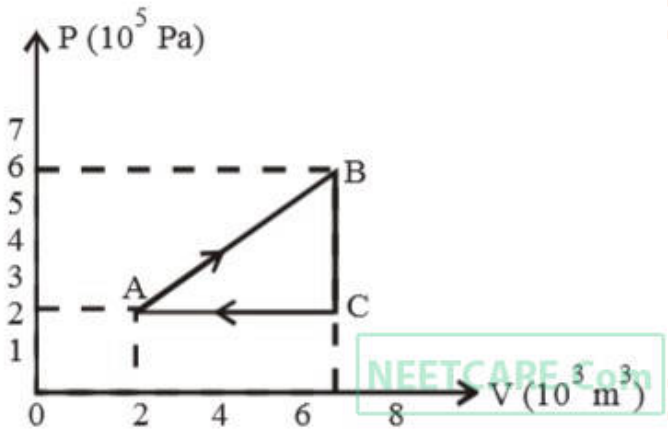


(A) $P_2 < P_1$
(B) Cannot be predicted
(C) $P_2 = P_1$
(D) $P_2 > P_1$

Q66. NEET 2013

A gas is taken through the cycle
A → B → C → A, as shown. What is the net work done by the gas?

Open solution



(A) Zero
(B) – 2000 J
(C) 2000 J
(D) 1000 J

Q67. NEET 2013

During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its temperature. The ratio of $\frac{C_p}{C_v}$ for the gas is

Open solution

(A) $\frac{5}{3}$
(B) $\frac{3}{2}$
(C) $\frac{4}{3}$
(D) 2

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Q68. NEET 2013



Open solution

The amount of heat energy required to raise the temperature of 1 g of Helium at NTP, from T_1K to T_2K is

(A) $\frac{3}{4}N_a k_B (T_2 - T_1)$

(B) $\frac{3}{4}N_a k_B \left(\frac{T_2}{T_1}\right)$

(C) $\frac{3}{8}N_a k_B (T_2 - T_1)$

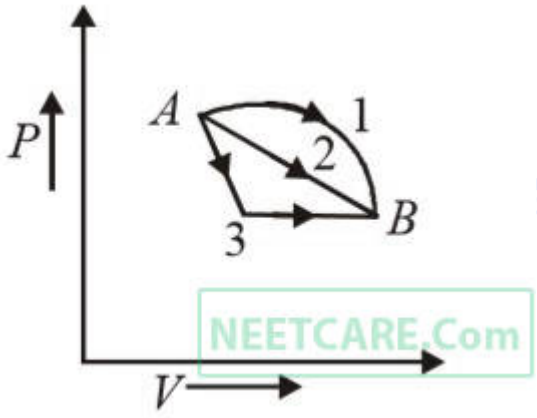
(D) $\frac{3}{2}N_a k_B (T_2 - T_1)$

Q69. AIPMT 2012 Mains



Open solution

An ideal gas goes from state A to state B via three different processes as indicated in the P-V diagram.



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If Q_1, Q_2, Q_3 indicate the heat absorbed by the gas along the three processes and ΔU_1, ΔU_2, ΔU_3 indicate the change in internal energy along the three processes respectively, then

(A) $Q_1 > Q_2 > Q_3$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$

(B) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$

(C) $Q_1 = Q_2 = Q_3$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$

(D) $Q_3 = Q_2 = Q_1$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$

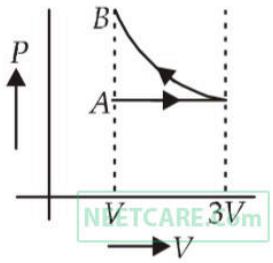
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Q70. AIPMT 2012 Prelims

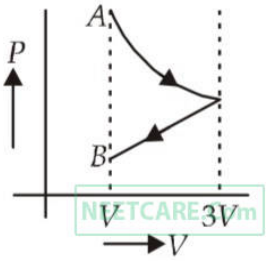
Open solution

One mole of an ideal gas goes from an initial state A to final state B via two processes : It first undergoes isothermal expansion from volume V to 3V and then its volume is reduced from 3V to V at constant pressure. The correct P-V diagram representing the two process is

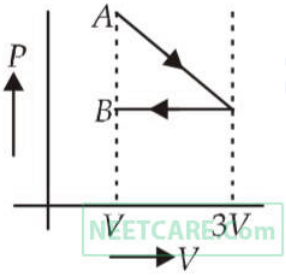
(A)



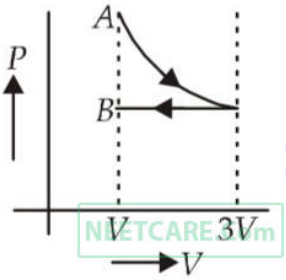
(B)



(C)



(D)

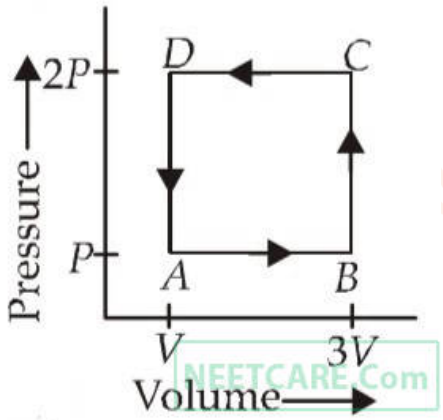


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Q71. AIPMT 2012 Prelims

A thermodynamic system is taken through the cycle ABCD as shown in figure. Heat rejected by the gas during the cycle is

Open solution



(A) 2PV (B) 4PV (C) $\frac{1}{2}$ PV (D) PV

Q72. AIPMT 2011 Mains

A mass of diatomic gas $(\gamma = 1.4)$ at a pressure of 2 atmospheres is compressed adiabatically so that its temperature rises from 27°C to 927°C. The pressure of the gas in the final state is

(A) 8 atm (B) 28 atm (C) 68.7 atm (D) 256 atm

Open solution

Q73. AIPMT 2011 Prelims

During an isothermal expansion, a confined ideal gas does $-$ 150 J of work against its surroundings. This implies that

(A) 150 J of heat has been removed from the gas

(B) 300 J of heat has been added to the gas

(C) no heat is transferred because the process is isothermal

(D) 150 J of heat has been added to the gas

Open solution

Q74. AIPMT 2011 Prelims

When 1 kg of ice at 0°C melts to water at 0°C, the resulting change in its entropy, taking latent heat of ice to be 80 cal/^{o}C, is

(A) 273 cal/K (B) 8×10^4 cal/K (C) 80 cal/K (D) 293 cal/K

Open solution

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Q75. AIPMT 2010 Mains

If c_p and c_v denote the specific heats (per unit mass of an ideal gas of molecular weight M, then

(A) $c_p - c_v = R/M^2$ (B) $c_p - c_v = R$ (C) $c_p - c_v = R/M$ (D) $c_p - c_v = MR$

Open solution

Q76. AIPMT 2010 Mains

A monatomic gas at pressure P_1 and volume V_1 is compressed adiabatically to $\frac{1}{8}^{th}$ of its original volume. What is the final pressure of the gas?

(A) $64P_1$ (B) P_1 (C) $16P_1$ (D) $32P_1$

Open solution

Q77. AIPMT 2010 Prelims

If ΔU and ΔW represent the increase in internal energy and work done by the system respectively in a thermodynamical process, which of the following is true ?

(A) $\Delta U = -\Delta W$, in an adiabtic process

(B) $\Delta U = \Delta W$, in an isothermal process

(C) $\Delta U = \Delta W$, in an adiabatic process

(D) $\Delta U = -\Delta W$, in an isothermal process

Open solution

Q78. AIPMT 2009

The internal energy change in a system that has absorbed 2 kcal of heat and done 500 J of work is

(A) 6400 J (B) 5400 J (C) 7900 J (D) 8900 J

Open solution

Q79. AIPMT 2009

In thermodynamic processes which of the following statements is not true ?

(A) In an isochoric process pressure remains constant.

(B) In an isothermal process the temperature remains constant.

(C) In an adiabatic process PV^γ = constant.

(D) In an adiabatic process the system is insulated from the surroundings.

Open solution

Q80. AIPMT 2008

At 10^oC the value of the density of a fixed mass of an ideal gas divided by its pressure is x. At 110^oC this ratio is

(A) $\frac{10}{110}x$ (B) $\frac{283}{383}x$ (C) x (D) $\frac{383}{283}x$

Open solution

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Q81. AIPMT 2008

If Q, E and W denote respectively the heat added, change in internal energy and the work done in a closed cyclic process, then

(A) E = 0 (B) Q = 0 (C) W = 0 (D) Q = W = 0

Open solution

Q82. AIPMT 2007

An engine has an efficiency of 1/6. When the temperature of sink is reduced by 62°C, its efficiency is doubled. Temperatures of the source is

(A) 37°C (B) 62°C (C) 99°C (D) 124°C.

Open solution

Q83. AIPMT 2006

A Carnot engine whose sink is at 300 K has an efficiency of 40%. By how much should the temperature of source be increased so as to increase its efficiency by 50% of original efficiency ?

(A) 380 K (B) 275 K (C) 325 K (D) 250 K

Open solution

Q84. AIPMT 2006

The molar specific heat at constant pressure of an ideal gas is (7/2) R. The ratio of specific heat at constant pressure to that at constant volume is

(A) 9/7 (B) 7/5 (C) 8/7 (D) 5/7

Open solution

Q85. AIPMT 2005

An ideal gas heat engine operates in Carnot cycle between 227°C and 127°C. It absorbs 6×10^4 cal of heat at higher temperature. Amount of heat converted to work is

(A) 4.8×10^4 cal (B) 6×10^4 cal

(C) 2.4×10^4 cal (D) 1.2×10^4 cal.

Open solution

Q86. AIPMT 2005

Which of the following processes is reversible?

(A) Transfer of heat by conduction (B) Transfer of heat by radiation

(C) Isothermal compression (D) Electrical heating of a nichrome wire.

Open solution

Q87. AIPMT 2004

The equation of state for 5 g of oxygen at a pressure P and temperature T, when occupying a volume V, will be
(where R is the gas constant)

(A) PV = (5/32)RT (B) PV = 5RT (C) PV = (5/2)RT (D) PV = (5/16) RT

Open solution

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Q88. AIPMT 2004

One mole of an ideal gas at an initial temperature of T K does 6R joule of work adiabatically. If the ratio of specific heats of this gas at constant pressure and at constant volume is 5/3, the final temperature of gas will be

(A) (T + 2.4) K (B) (T − 2.4) K (C) (T + 4) K (D) (T − 4) K

Open solution

Q89. AIPMT 2003

An ideal gas heat engine operates in a Carnot cycle between 227°C and 127°C. It absorbs 6 kcal at the higher temperature. The amount of heat (in kcal) converted into work is equal to

(A) 4.8 (B) 3.5 (C) 1.6 (D) 1.2

Open solution

Q90. AIPMT 2002

The efficiency of Carnot engine is 50% and temperature of sink is 500 K. If temperature of source is kept constant and its efficiency raised to 60%, then the required temperature of sink will be

(A) 100 K (B) 600 K (C) 400 K (D) 500 K

Open solution

Q91. AIPMT 2001

A scientist says that the efficiency of his heat engine which work at source temperature 127°C and sink temperature 27°C is 26%, then

(A) it is impossible
(B) it is possible but less probable
(C) it is quite probable
(D) data are incomplete.

Open solution

Q92. AIPMT 2000

The (W/Q) of a Carnot engine is 1/6, now the temperature of sink is reduced by 62°C, then this ratio becomes twice, therefore the initial temperature of the sink and source are respectively

(A) 33°C, 67°C (B) 37°C, 99°C (C) 67°C, 33°C (D) 97 K, 37 K

Open solution

Q93. AIPMT 2000

To find out degree of freedom, the expansion is

(A) $f = \frac{2}{\gamma - 1}$ (B) $f = \frac{\gamma + 1}{2}$ (C) $f = \frac{2}{\gamma + 1}$ (D) $f = \frac{1}{\gamma + 1}$

Open solution

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Answer Key

1. **D**	2. **D**	3. **B**	4. **A**	5. **D**
6. **A**	7. **D**	8. **A**	9. **D**	10. **A**
11. **D**	12. **B**	13. **D**	14. **C**	15. **A**
16. **D**	17. **D**	18. **D**	19. **A**	20. **D**
21. **D**	22. **A**	23. **D**	24. **B**	25. **D**
26. **C**	27. **D**	28. **A**	29. **A**	30. **A**
31. **D**	32. **A**	33. **A**	34. **A**	35. **C**
36. **C**	37. **B**	38. **A**	39. **A**	40. **B**
41. **D**	42. **A**	43. **A**	44. **C**	45. **D**
46. **B**	47. **B**	48. **D**	49. **D**	50. **A**
51. **C**	52. **D**	53. **C**	54. **A**	55. **A**
56. **D**	57. **A**	58. **C**	59. **D**	60. **B**
61. **D**	62. **D**	63. **C**	64. **A**	65. **A**
66. **D**	67. **B**	68. **C**	69. **A**	70. **D**
71. **A**	72. **D**	73. **D**	74. **D**	75. **C**
76. **D**	77. **A**	78. **C**	79. **A**	80. **B**
81. **A**	82. **C**	83. **D**	84. **B**	85. **D**
86. **C**	87. **A**	88. **D**	89. **D**	90. **C**
91. **A**	92. **B**	93. **A**		

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